Ahmed El-Hajjar

Technical Writer, Software Engineer

https://www.cdahmedeh.net • linkedin.cdahmedeh.net • github.cdahmedeh.net

cdahmedeh@gmail.com • (613) 800-9008 • Ottawa, ON, Canada

Leadership and Initiatives

Ottawa Computing Group – meetup.com/ottawa-computing-group

***Founder and Host** • 300+ events • 1300+ members • 6+ years* June 2019 – Current

Weekly in-person social club spanning technology, current affairs, project showcases and free-form discussions.

- Plan, coordinate and moderate one of the few technology meetups that remained fully in-person after the pandemic.
- Attendees share projects, challenge each other's perspectives, and occasionally discover job opportunities or collaborators.
- No presentations, agendas, speakers, or predetermined topics. Conversations are entirely driven by attendees.
- Welcomes everyone from software developers, blue-collar experts, content creators, entrepreneurs and even non-techies.

Selected Portfolio Projects

PoetWrite – github.com/cdahmedeh/PoetWrite

Technical Writer, Software Engineer, Architect Solo Open-Source, Early Development, Documentation Focused, Java

Research-phase word processor for poets featuring rhetorical analysis and lexicographic assistance. Fully documented software architecture and analyses engines.

- Comprehensively documented custom asynchronous MVVM UI framework using reactive patterns, bespoke caching systems and a clever lazy-loading dependency-injected services.
- Research how far deterministic, AI-free language analysis and assistance can be pushed before requiring machine learning techniques.
- Sole ownership of project using software engineering practices. Such as architecture design, UX design, software development, documentation, quality assurance and TDD-driven testing.

Crosswind – github.com/cdahmedeh/Crosswind

Technical Writer, Reverse-Engineer, Programmer Solo Open-Source, Prototype Release, .NET C#

Reverse-engineering effort to create compatibility layer that bridges modern flight planning software to decades-old legacy flight simulators.

- Crosswind allows users of older flight simulators to use modern flight planning tools that don't have published APIs for providing flight or environment data or access to aircraft analytics.
- Clean-room reversed-engineered undocumented protocols using packet sniffers and injected API monitors. To develop a data translation layer between legacy simulator interfaces (FSUIPC) and modern flight-planning APIs (Navigraph Simlink).
- Interfaces and protocol findings documented in source code along with data structures used by said protocols.
- Distributed as a complete open-source deliverable. With release notes, compatibility lists, build instructions, usage instructions in the repository home page. Included is a fully compiled distributable with changelog, download links and hashes for validation.

TopRoms – github.com/cdahmedeh/TopRoms

***Technical Writer, Curator, Infrastructure** • 176 GitHub Stars • 4000+ titles • 60+ platforms • Since 2017* Solo, Fully Released

Curated preservation project focused on historically significant and high-quality retro games. Designed as a more practical alternative to full-archival collections.

- Research-driven curation methodology combining review aggregation, historical sources, community feedback.
- Authored extensive project documentation covering collection history, curation methodology, emulator recommendations, platform compatibility, release notes, and setup guidance.

- Built and maintain a large-scale ROM content-delivery project end-to-end. Maintaining hosting and distribution infrastructure as a seedbox. Conversion of formats for emulator compatibility using tools from the MAME project.

Selected Publications

Automatic Transmission Simulation in Games – automatic.cdahmedeh.net

Technical deep-dive comparing accuracy of automatic transmission simulation across driving games.

- Results from research project investigating why the majority of driving games have inaccurate modeling of automatic transmissions. And describe, test and demonstrate findings of rare and obscure titles that accurately reproduced the behaviour of automatics found in road cars.
- Explains the engineering principles behind automatic transmission behaviour in a manner accessible to driving game enthusiasts.
- Develop a testing methodology to identify the various distinctive characteristics and features for the in-game analyses.
- Includes comparative testing and video demonstrations documenting these tests across simulators and driving games.
- Widely referenced by driving simulation enthusiasts and developers as a resource on transmission modeling.

The Sad Demise of Propulsion Controlled Aircraft – pca.cdahmedeh.net

Storytelling and historical technical case study of propulsion-controlled aircraft and its impact on aviation safety and its eventual cancellation and abandonment.

- Researched a little-known NASA project that allowed pilots to control and land aircraft with complete hydraulic and flight control failure using engine thrust alone.
- Connected historical accident reports, NASA and McDonnell-Douglas flight-test programs, aviation safety research, and simulator demonstrations through storytelling.
- Supported with videos of CVR recordings, accident recordings, flight-test program initiatives and results, and reference to a simulator demo demonstrating throttle-only flight control.
- Reflection on how FAA considerations influenced the rejection of a technology that may have prevented several major aviation disasters.

Professional Experience

Flybits

Technical Writer, Solutions Team March 2024 – June 2026

Owner and author of documentation ecosystem for complex fintech platform that had major knowledge fragmentation challenges. Serving engineering and developer documentation, comprehensive product user guides, cross-departmental knowledge transfer, support enablement and professional services.

Creating a from-scratch comprehensive SDK development reference documentation and tutorials from fragmented sources.

- Flybits was a customer-engagement platform where marketers created and managed personalized content experiences through a web application, while the SDK delivered those experiences to mobile and web applications.
- SDK knowledge was fragmented across outdated documentation, engineering tribal knowledge, support teams, and professional services.
- Lack of accurate reference material increased support burden and made developer onboarding difficult.
- Collected information through SME interviews and knowledge-transfer sessions with engineering, product, solutions, and professional services teams.
- Supplemented code examples and snippets through SDK source-code analysis, internal demo applications, and reverse-engineering of undocumented library functions.
- Developed prototype applications in Swift and Kotlin to validate documentation accuracy and generate executable code samples, allowing documentation to be written from direct implementation experience rather than theory alone.

- Designed documentation structure for searchability and discoverability through SEO-like search optimization and metadata tagging.
- The majority of the documentation was written from scratch via the GitBook platform. Used and a combination of reference-style and step-by-step writing patterns tailored to both novice and experienced developers.
- Structured documentation into a progressive learning path combining tutorials, implementation guides, troubleshooting references, API configuration guides, and platform-specific Android and iOS documentation.
- Developed and maintained a multi-hundred-page SDK documentation system covering the full developer lifecycle from initial setup and authentication through testing, deployment, and advanced platform capabilities.
- Kept and maintained older documentation for legacy and posterity, as some customers were still transitioning to newer frameworks.

Designing and composing User Guide for primary web application platform for client end-users and marketers.

- Flybits operated a web application, called Experience Studio, used to design and manage personalized content experiences across mobile and web applications, including contextual content presentation, audience targeting, scheduling, prioritization, and analytics.
- Primary audiences had limited technical backgrounds, including marketers and product owners. Documentation balanced ease of use for business stakeholders with the technical accuracy required for engineering collaboration.
- Platform development was fast-paced, with major UX and workflow changes occurring frequently throughout the product's evolution and redesign efforts.
- Virtually no documentation existed for the redesigned platform, with available material limited to deprecated documentation from earlier generations of the product.
- Developed a strategy for creating a comprehensive user guide from scratch, working closely with product, support, and solutions teams to ensure documentation accurately reflected platform behavior and customer needs.
- Documentation structure was inspired by large-scale enterprise software user guides and the progressive learning approach found in flight operating manuals.
- Created a gradual learning path that introduced concepts in concise stages before progressing to workflows, tutorials, and practical examples, rather than relying primarily on reference-style paradigms.
- Support and solutions teams reported substantially less time spent explaining platform fundamentals, allowing greater focus on developing customer-specific solutions and implementation guidance.
- Recognized that many end-users and marketers are reluctant to read documentation, leading to a simplified structure with extensive visual aids, animations, strong searchability, and step-by-step explanations and navigation.
- Implemented a tagging system to track documentation coverage of platform features, as a management strategy as the platform evolved and new features were implemented.

Writing onboarding documentation for finance and accounting team.

- Flybits underwent a major transformation of its finance operations, including billing, reimbursement, payroll, and accounting workflows, accompanied by migration to entirely new platforms and processes.
- The redesigned processes were primarily understood by the accounting leads, who required comprehensive onboarding documentation to train and support the rest of the finance team.
- Had no prior experience with finance or accounting systems and was required to rapidly learn unfamiliar concepts and terminology under tight project timelines due to the urgency of the documentation effort.
- Structured documentation around financial activities and workflows rather than linear beginning-to-end narration, allowing team members to learn and reference individual responsibilities independently.
- Authored documentation in Google Docs rather than developer-oriented documentation platforms to align with the limited technical-familiarity and working practices of the finance team.
- Due to confidentiality restrictions, direct access to financial systems was unavailable. Documentation was created through detailed note-taking, meeting transcriptions, process mapping, copious screenshots, and frequent and long SME interviews.

Syntronic

Technical Writer, Business Analyst May 2022 – November 2023

Requirements analyst and technical communicator for a variety of research initiatives, including medical labs, embedded systems and software development. Additional specializations into turning fragmented knowledge into concise documentation in a multitude of projects. Consolidated with knowledge-transfer sessions, collaboration with engineering teams, and product. Restructuring documentation and formal readability analysis.

- Conducted client interviews and requirements-gathering sessions to determine the operational and technical needs of a newly developed automated medical sterilization laboratory.
- Analyzed, validated, and refined requirements by identifying ambiguities, contradictions, missing information, and implementation risks through collaboration with clients, analysts, and subject-matter experts.
- Transformed business and operational requirements into software specifications, implementation plans, and development roadmaps for future software engineering initiatives.
- Authored user stories, acceptance criteria, and test scenarios to support development and quality-assurance activities.
- Collaborated with developers and UX specialists to formalize workflows, use cases, and functional requirements prior to implementation.
- Contributed to business-analysis reports documenting project objectives, development plans, assumptions, risks, and findings discovered throughout the requirements-engineering process.

Automating requirements management with continuous integration and deployment platforms.

- Requirements, specifications, and implementation plans were maintained primarily in Microsoft Word and PDF documents, resulting in significant manual effort to transfer information into Azure DevOps.
- Developed an application to automatically extract requirements and acceptance criteria from specification documents.
- Integrated with Azure DevOps APIs to automatically synchronize the above items to eliminate the need for manual data entry thus improving consistency between specifications and project tracking.

Technical communications for emerging cloud and embedded technologies.

- Authored whitepapers, brochures and marketing material describing digital-transformation initiatives, and embedded-system products for prospective clients.

Early Career

Amazon, Entrust, 1VALET

Technical Writer, Consultant October 2020 – October 2022

Focused on complex systems that were poorly documented. For robotics platforms, automation infrastructure, web security tools, property technology. Looking into source code, interviews with SMEs, and tracing actual workflows rather than just theoretical test cases. With that, produce onboarding guides, user docs, success guides, knowledge-transfer documentation. Targeted both for engineers and for support who don't have much technical fluency.

Cisco, ESDC Service Canada, Ford Motor Company

Software Consultant February 2019 – October 2020

Consultation for modernization of legacy enterprise systems for modern platforms. Software engineering, automation, deployment, testing, and architecture initiatives. In telecommunications, government, and automotive sectors. Documentation for cloud infrastructure and deployment, continuous integration and deployment, automated testing, technical documentation, containerization.

Lixar, NorthStar, ShopLiftr, MediaVantage, Teldio

Software Developer December 2013 – February 2019

Full-stack software developer authoring backend cloud platforms, APIs, mobile applications, enterprise applications, machine learning, IoT. And research and concepts for emerging technologies. Responsible for architecture, backend development, DevOps, documentation, product support. Servant leadership and mentoring junior developers.

Skills

Domains and Industries

Financial Technologies, Healthcare, Medical Technology, Laboratory Automation, Robotics, Cybersecurity, Telecommunications, Networking, VoIP, Property Technology, Government, Public Sector, Automotive, Retail Technology, Marketing Technology, Utilities, Cloud Services, Enterprise Software, Mobile Applications, Web Applications, Modeling and Simulation

Documentation and Knowledge Management

Developer Documentation, SDK Documentation, API Documentation, Technical Documentation, User Guides, Success Guides, Deployment Documentation, Support Documentation, Knowledge Transfer, Release Notes, Changelogs, Version History, Specifications, Requirements, User Stories, Test Cases, Prototyping, Information Architecture, Technical Communication

Emerging Technology and Research

Artificial Intelligence, Machine Learning, OCR, Blockchain, Bluetooth, Proximity Technologies, Software-Defined Radio

Business Analysis and Product Discovery

Requirements Gathering, Requirements Engineering, Specifications Development, Customer Interviews, SME Interviews, Business Analysis, Use Cases, Workflow Analysis, Process Mapping, Acceptance Criteria, Product Ownership, Product Support, Code Exploration, Visualization

Leadership and Collaboration

Project Management, Stakeholder Management, Interdepartmental Communication, Mentoring, Peer Reviews, Servant Leadership, Event Hosting, Moderation, Knowledge Transfer

Visual Modeling and Design

UML, Sequence Diagrams, State Diagrams, Flowcharts, Swim Lanes, Mind Maps, Wireframing, Balsamiq, Microsoft Visio, draw.io, User Experience (UX)

Content Creation and Publishing

Whitepapers, Case Studies, Blogging, Ghostwriting, Copywriting, Editing, Research, Fact Checking, Marketing Communications, Announcements, Social Media, Content Strategy

Documentation and Authoring Tools

GitBook, Confluence, MediaWiki, SharePoint, Google Docs, Microsoft Word, Scriviner, Markdown, Antidote

Programming Languages

Java, C#, Python, JavaScript, Swift, Kotlin, C++, Go, PHP, Ruby, Visual Basic

Scripting Languages

bash, PowerShell, AppleScript

Version Control and Source Control

Git, GitHub, GitLab, Bitbucket, SVN, CVS, Branching, Merging, Tagging, Pull Requests, Code Reviews, Release Management

Web Development

HTML, CSS, JavaScript, Angular, React, Vue.js, Single Page Applications, Server-Side Rendering

Software Development and Architecture

Object-Oriented Design, Functional Programming, MVC, MVVM, Dependency Injection, Multi-Threading, Asynchronous Programming, Reactive Programming, Reverse Engineering, Test-Driven Development, Software Architecture

Mobile Development

Android, iOS, Firebase, APNs, Retrofit, Mobile SDK Integration

Backend Development

Java EE, Spring, Spring Boot, Django, Ruby on Rails, REST APIs, SOAP, Swagger, Postman

Database and Data

SQL, PostgreSQL, MySQL, SQL Server, Oracle, SQLite, Hibernate, JPA, Liquibase, Flyway

DevOps and Platforms

Azure DevOps, GitHub, GitLab, Bitbucket, Jenkins, TeamCity, Bamboo, Docker, Docker Compose, CI/CD, Maven, Gradle

Operating Systems and System Administration

Windows, Linux, macOS, SSH, Windows Server, Package Management, System Administration, Virtualization, Backup Systems

Cloud and Infrastructure

Amazon AWS, Microsoft Azure, DigitalOcean, S3, CloudWatch, Redshift, DynamoDB

Embedded and Hardware

Embedded Systems, FPGA, Arduino, Raspberry Pi, Circuit Design, PCB Components, Soldering, 3D Printing

Spoken Languages

English, French, Arabic, Cat (Domestic Fluency)

Additional Experience

Uber

Partner and Driver *• Part-Time • 4.9 Star Rating • 1000+ trips* January 2019 – September 2025

Independent customer-service and transportation operation serving riders throughout Ottawa-Gatineau while managing scheduling, bookkeeping, and day-to-day business operations. Helped develop communication and interpersonal skills that could be applied in other areas in my field.

Volunteering

Ottawa Stray Cat Rescue

Lead Volunteer *• Occasional* February 2020 – Current

Organizing fundraisers and adoption events for the local animal rescue for stray and feral cats throughout Ottawa.

Selected Essays

Electric Youth -- Marnie – marnie.cdahmedeh.net

Analytical essay exploring music theory, lyrical themes, visual storytelling, and artistic intent within a contemporary synth-pop composition.

Education

Studies in Software Engineering, Computer Engineering, Computer Science

University of Ottawa September 2009 – December 2014

Studies in Technical Communications

Simon Fraser University (Online) June 2018 – November 2018